

The Categorical Structure of Federated Convergence: Limits, Sheaves, and Minimal Coordination

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Abstract

Normalization confluence gives coordination-free convergence within a single registry, and its federated extension gives convergence across a network of registries connected by morphisms, under two decidable regimes: acyclic networks, and monotone repair on cycles [1, 2]. Those regimes are stated as separate sufficient conditions. This paper gives the structure underneath them and, with it, an answer to a sharper question: when must a federation coordinate at all, and how little coordination suffices?

We show that a federation’s convergent states are a limit and its federated normalizer is the retraction onto that limit, so compositionality (a verified sub-federation collapses to an effective registry) is a corollary of one structural fact rather than a per-topology argument. We then show that convergence certificates form a sheaf whose gluing condition is exactly the source-determinacy and validity-preservation hypotheses the federation already certifies, so the obstruction to a global convergent state is cohomological: H^0 is the convergent states and H^1 is the holonomy of the morphism cycles. The two convergence regimes are then one picture: an acyclic network has no cycles and H^1 vanishes, while a monotone cycle collapses the obstruction to a least fixed point. The obstruction is computable as a loop composite over the shared subspace, implemented in the reference engine as a cycle diagnostic that names an obstructing cycle and exhibits its orbit. Finally we show that when the obstruction is nonzero, coordinating a cycle basis of the morphism network suffices to converge, localizing the global coordination verdict of CALM and invariant confluence into a mixed-consistency partition. The exact minimum coordination is the group feedback edge set number of the labeled nerve: NP-hard even in the abelian case, but fixed-parameter tractable in the size of the coordinated core and polynomial on planar or edge-disjoint nerves. The structural core (the limit, the retraction, compositionality, and order-independence) and the cohomological layer (the completion on arbitrary graphs, the cycle-basis criterion, H^1 as a quotient with its rank, and the S_3 separation) are mechanized axiom-free in Coq/Rocq.

Keywords: Federated Convergence, Normalization Confluence, Sheaf Cohomology, Coordination Avoidance, CALM, Category Theory, Limits, Holonomy, Minimal Coordination, Mechanized Proof

1 Introduction

Normalization confluence certifies that a single registry converges: under well-founded compensation and compensation commutativity, every processor that consumes the same events reaches the same valid state regardless of order [1]. Its federated extension certifies that a network of such registries, connected by directed morphisms that carry one registry’s authority over another’s shared state, converges as a whole, in two regimes: any acyclic network whose morphisms preserve validity, and any network (cyclic included) whose repair is monotone [2]. Both regimes are established by finite enumeration and are what the reference engine checks at build time.

Those two regimes are presented as independent sufficient conditions, proved by separate arguments. That leaves a structural question unanswered and a practical one unasked. The structural question: is there one reason both regimes work, rather than two? The practical question: when a federation falls into neither regime, the engine rejects it, but rejection is a blunt verdict. A designer wants to know not merely that the network cannot converge coordination-free, but *where* the conflict is and *how little* coordination would repair it. This paper answers both, and the answers turn out to be the same object viewed twice.

The local-to-global view. A federation poses a local-to-global problem. Each registry is convergent on its own; each morphism, in isolation, is a clean rule for propagating authority. The question is whether, once every morphism constraint holds at once, there is a single global assignment the whole network agrees on, and whether it is reached regardless of event order. This is the shape of a limit, and of a sheaf gluing problem, and the obstruction to it is the shape of a cohomology class. Making that precise is what unifies the regimes: an acyclic network has no cycles to obstruct gluing, and a monotone cycle has an order that forces gluing even when a cycle is present. The same H^1 that measures the obstruction also, when it is nonzero, pinpoints the cycles responsible, which is exactly the diagnostic and the minimal-coordination answers the practical question wants.

Scope, stated up front. The cleanest form of the cohomological classification holds in the *invertible* fragment, where each morphism transports its shared value by a bijection (a lossless relabeling: a copy, a permutation, a bijective unit or schema map). There the transitions are group elements and H^1 is genuine Čech cohomology; a cycle basis is a sufficient coordinated core, though computing the minimum one is NP-hard even here (Section 9). When morphisms collapse information (a clamp, a reset, a most-restrictive-wins merge, the shape most compensation takes), the transitions are monoid elements rather than group elements, classical H^1 does not apply, and the correct statement is the operational one: a global convergent state exists iff the loop composite of each cycle has a reachable fixed point. We therefore lead with the general, operational results (the limit, the retraction, compositionality, and the loop-composite diagnostic), which hold for arbitrary morphisms and are what the engine implements, and present the cohomology as the sharp closed form on the invertible fragment. We state which real federations inhabit that fragment (Section 9) rather than leaving it as “the case where the math is clean.”

Contributions.

1. **Federation as a limit, and compositionality for free (Sections 4–5).** A registry’s valid set is the equalizer of its idempotent normalizer, a federation’s convergent set is the equalizer of its shared-projection and resolver maps, and the federated normalizer is the

idempotent retraction onto that limit. Because the normal forms are a limit, a convergent sub-federation collapses to an effective registry and the combined normalizer factors through it; compositionality is a corollary of one structural fact, not a per-topology theorem.

2. **Convergence certificates form a sheaf, and $R1/R2$ is its gluing axiom (Section 6).** Local convergence certificates glue to a global one exactly when the subsystems agree as normalizers on their overlap. The two regimes that force agreement, single-writer authority and monotone overlap, are the federation’s source-determinacy ($R1$) and validity-preservation ($R2$) hypotheses. The existing federation design is validated as the sheaf gluing check, not extended by a new mechanism.
3. **The obstruction is cohomological, and it unifies the two regimes (Section 7).** H^0 is the convergent states; H^1 (in the invertible fragment) is the holonomy of the morphism cycles, generated by a cycle basis. H^1 vanishes in exactly two ways, and they are the two convergence regimes: an acyclic network (no cycles) and a monotone cycle (a least fixed point). The dichotomy the base work states as two conditions is one invariant.
4. **A computable obstruction diagnostic (Section 8).** For a general (possibly non-invertible) cycle, a global convergent state exists iff the cycle’s loop composite has a reachable fixed point, and the witness for its absence is the loop composite itself, computable over the shared subspace rather than the product. This is implemented in the reference engine as a cycle diagnostic that names the offending cycle and reports whether its repair settles or orbits.
5. **Minimal coordination (Section 9).** When the obstruction is nonzero, coordinating a cycle basis of the morphism network trivializes it, yielding a convergent implementation that is strongly consistent only on those loops and coordination-free everywhere else. This refines the global coordination verdict of CALM [5, 6] and invariant confluence [7] into a localized, mixed-consistency partition, and it upgrades the engine’s blunt rejection of a nonconvergent cycle into an actionable “coordinate these loops.” We identify the exact minimum with the group feedback edge set number of the labeled nerve, resolving its complexity: NP-hard even in the abelian case (it contains edge bipartization), fixed-parameter tractable in the size of the coordinated core, and polynomial on planar or edge-disjoint nerves; and we show the minimum is never below the abelian count and can be strictly above it, so abelianized sizing is unsound.
6. **Mechanization.** The structural core (Contributions 1 and 2: the limit, the retraction for arbitrary acyclic federations, and compositionality) is mechanized axiom-free in Coq/Rocq, extending the machine-checked development of the base results, together with full order-independence over arbitrary acyclic federations. So is the cohomological layer: the completion theorem on an arbitrary group-labeled graph, the cycle-basis criterion, the sufficiency of coordinating a spanning tree’s complement, H^1 as tuples of fundamental holonomies modulo simultaneous conjugation with rank $|E| - |V| + 1$, and the S_3 separation including the graph-level counts $\min_G = 2 > 1 = \min_{G^{\text{ab}}}$.

2 Related Work

Sheaf cohomology for distributed systems: the same tool, a different object. The use of sheaf cohomology to measure a local-to-global obstruction is not new, and we do not claim it. Cellular-sheaf cohomology characterizes solvability of distributed decision *tasks*, where terminating solutions are the global sections and the cohomology encodes the obstruction to a solution [9], in the combinatorial-topology lineage of distributed computing [10]; the same

presheaf-and-obstruction template measures quantum contextuality [8]. Our object is different. Those works concern which one-shot decision task a set of processes can solve under a failure or adversary model. We concern the convergence of a *federated normalization*: a network of registries, each with invariants and compensation, whose morphisms must agree on shared state so that all processors reach the same valid global normal form regardless of event order. Our presheaf is over the overlaps of a normalizer structure, its sections are convergence certificates, H^0 is the convergent states, and H^1 is the holonomy of the morphism cycles. To our knowledge no prior sheaf-cohomology work targets the eventual-consistency or replicated-convergence obstruction, and none yields a coordination-minimality result.

Coordination avoidance: a global verdict we localize. The CALM theorem states that a problem has a coordination-free, consistent implementation iff it is monotone [5, 6], and invariant confluence [7] gives a related global avoidability criterion; recent work sharpens the monotonicity boundary at the semantic level [11, 12]. These answer *whether* coordination can be avoided, as a global property of the whole problem. Our minimal-coordination result answers *where* and *how little*: it locates the irreducible coordination in a cycle basis of the morphism network and shows the rest can remain coordination-free, a localized mixed-consistency partition rather than a single yes-or-no. In the information-preserving fragment the core is provably minimal. We are not aware of prior work that localizes the coordination requirement to a cycle basis or bounds it by a Betti number.

Convergent replicated data and its algebra. Conflict-free replicated data types achieve convergence by restricting to operations that never need repair; a state-based CRDT is the join-semilattice special case [4], and the monotone regime here coincides with it (this correspondence is machine-checked in the base development). The well-founded-compensation regime is strictly broader, since it admits operations that violate invariants and are repaired. The correspondence also holds against the literature’s op-based definition, where only concurrent operations commute under causal delivery: the base development proves convergence when governed steps commute on concurrent pairs only (`causal_convergence`, with standard op-based CRDTs as the instance `causal_cmrdt_SEC`), shows the compensation-free fragment is exactly the op-based CRDTs (`compensation_free_exact`, both directions), keeps the inclusion strict (`witness_causal_not_cmrdt`), and proves the rewrite-system convergence theorem `causal_governance_confluent` with the critical-pair condition required only for co-enabled distinct events, all axiom-free. Category-theoretic accounts of CRDT convergence exist at the level of the merge algebra; they do not give a cohomological obstruction or a coordination-minimality theorem. The lattice-theoretic and chaotic-iteration machinery behind the monotone regime is classical [13, 14], as is the confluence core [3].

Invariant repair and mechanized convergence. The base development sits in the invariant-repair lineage: Indigo [15] and IPA [16] also restore invariants over weakly consistent replicas, by reservation and invariant-confluence-preserving analysis rather than by confluence of a compensating rewrite system. Its method roots are Meseguer’s rewriting logic and coherence [17, 18] and self-stabilization [19]. On the mechanization side, Gomes et al. [20] machine-check strong eventual consistency for CRDTs in Isabelle/HOL; the base convergence theorem over which we build the categorical structure is likewise machine-checked and axiom-free, for the broader compensation regime. None of these give a cohomological obstruction or a coordination-minimality theorem, which is the object of this paper.

3 Background

We restate only what later sections use; full statements and proofs are in the base papers [1, 2].

Single registry. A registry governs a finite state space by a validity predicate and a compensation (repair) map. Its normalizer ρ_R iterates compensation to a valid state. Two conditions make this well-behaved: *well-founded compensation* (WFC), a compensation measure that strictly decreases on every invalid state, so compensation terminates and ρ_R is total; and *compensation commutativity* (CC), the local-confluence condition on event and repair steps. Under WFC and CC, Newman’s Lemma [3] gives unique normal forms: every processor that consumes the same events reaches the same valid state regardless of order. WFC also makes ρ_R idempotent (a normal form is a fixed point) and its valid set non-empty (compensation reaches a valid state from any state), the two facts Section 4 uses.

Federation. Registries are connected by directed *morphisms*. A morphism $\phi : A \rightarrow B$ makes A authoritative over a designated *shared* component of B : given A ’s normal form, ϕ fixes B ’s shared variables, leaving B ’s local variables to B ’s own compensation. The core mechanism is the *authority argument*: because the source has a unique normal form, it deterministically fixes the target’s shared component, eliminating the nondeterminism a symmetric constraint would introduce. A single-source morphism must preserve validity under overwrite ($M1$): overwriting a valid target’s shared component with the morphism image of a valid source keeps the target valid. A target with several sources carries a *resolver* that merges them, subject to source-determinacy ($R1$: it reads only the sources) and validity preservation ($R2$: the single-source $M1$ generalized). The base work proves federated convergence in two regimes, each by finite enumeration: any acyclic network whose morphisms satisfy $M1/R1/R2$, and any network (cycles included) whose repair is monotone (least fixed point by Knaster-Tarski [13], reached by chaotic iteration [14]). Those two regimes are what this paper explains as one structure.

4 Normal Forms Are a Limit

We work at the semantic level. A registry R has a finite state space D_R and a *normalizer* $\rho_R : D_R \rightarrow D_R$, the iterated-compensation map, which is total (termination, from well-founded compensation) and idempotent ($\rho_R \circ \rho_R = \rho_R$: a normal form is a fixed point, from confluence plus termination). Write $\Phi_R := \text{im}(\rho_R)$ for the valid set, the normal forms.

Everything in this section is elementary in **Set**, and the load-bearing facts are mechanized axiom-free in Coq/Rocq (module `Categorical.v`); we state them and sketch the arguments, naming the mechanized lemmas.

4.1 A single registry is an equalizer

Lemma 4.1 (Lemma 0). *For an idempotent normalizer ρ_R , the valid set, the fixed-point set, and the equalizer of the identity and ρ_R coincide:*

$$\Phi_R = \text{im}(\rho_R) = \text{Fix}(\rho_R) = \text{eq}(\text{id}_{D_R}, \rho_R),$$

a limit in Set.

Proof. Idempotence gives $\text{im}(\rho_R) = \text{Fix}(\rho_R)$: if $s = \rho_R(x)$ then $\rho_R(s) = \rho_R(\rho_R(x)) = \rho_R(x) = s$, and conversely a fixed point is its own image. And $\text{Fix}(\rho_R) = \{s : \text{id}(s) = \rho_R(s)\}$ is by definition the equalizer of id and ρ_R , a limit. Mechanized as `image_iff_fixed` and `fixed_is_equalizer`. $\square$

Read as a retraction onto Φ_R (it fixes Φ_R and lands in it), ρ_R is an idempotent monad on **Set** whose algebras are the normal forms, and a closure operator on the ordered domains of the cyclic regime. We need only the equalizer.

4.2 A federation is a finite limit

Fix a federation: registries $R_1, \dots, R_n$, and for each target B with source set $\text{src}(B)$, shared variables $\text{sh}(B)$ and a resolver Res_B merging its sources' normal forms into B 's shared component (for a single source this is the morphism's map). The base papers certify by finite enumeration that each Res_B reads only its sources (*R1*) and preserves target validity (*R2*; for a single source, *M1*). Let g send a tuple of component normal forms to the tuple of the targets' current shared components, and h send it to the tuple of resolver values. The *consistent set* is

$$L_F := \left\{ (s_1, \dots, s_n) \in \prod_i \Phi_{R_i} : \text{for every target } B, s_B \upharpoonright \text{sh}(B) = \text{Res}_B((s_A)_{A \in \text{src}(B)}) \right\}.$$

Proposition 4.2 (Proposition 1). $L_F = \text{eq}(g, h)$. Since each Φ_{R_i} is a limit (Lemma 4.1) and L_F is an equalizer of a product of these, L_F is a finite limit in **Set**.

Proof. A tuple lies in L_F iff every target's shared component equals its resolver value, that is, iff g and h agree on it componentwise, that is, iff $g = h$ there. Componentwise agreement over the finite target set is equality of the two tuple-valued maps, so L_F is their equalizer. Mechanized as `consistent_iff_equalizer`, with the product over targets modeled as a list so the characterization needs no functional extensionality. $\square$

4.3 The federated normalizer is the retraction onto the limit

Fix a topological order (sources before targets). The federated normalizer ρ_F processes registries in that order: for each, overwrite its shared component with the resolver applied to the already-finalized sources, then apply the local normalizer. We model this as a left-to-right fold over the ordered components, writing `step` for the per-position operation (read finalized prefix, overwrite shared, normalize). The one hypothesis it needs is that `step` is idempotent given a fixed finalized prefix, which is exactly *M1/R2* (the local normalizer fixes the overwritten shared component) together with local idempotence.

Theorem 4.3 (Theorem 1: federation retraction, acyclic). ρ_F is an idempotent retraction onto L_F : $\text{im}(\rho_F) = \text{Fix}(\rho_F) = L_F$. Equivalently, the federated normal forms are exactly the finite limit of Proposition 4.2 and ρ_F is the retraction onto it. Moreover ρ_F is independent of the chosen topological order.

Proof sketch. Two facts suffice and yield the retraction by a general argument: any operator that is sound and complete for a set L is the idempotent retraction onto L , with image and fixed set L (mechanized as `RetractionOntoConsistent`). *Soundness* (Lemma A): $\rho_F(s) \in L_F$, because each finalized position is step of its finalized prefix, which step-idempotence leaves fixed (`rhoFold_sound`). *Completeness* (Lemma B): a consistent state is already fixed,

since each position's step is a no-op on it; this uses only the definition of L_F , not $M1/R2$ (`rhoFold_complete`). Hence ρ_F is the idempotent retraction onto L_F (`rhoFold_retraction`, `rhoFold_image_iff_consistent`), for arbitrary acyclic federations, not a fixed small instance.

Order-independence reduces to the commutation of two incomparable registries: their reads and writes are disjoint, so processing them in either order gives the same result (`updates_commute`). Lifting this to all topological orders classically uses the connectivity of a finite poset's linear extensions under adjacent transpositions of incomparable elements. The mechanization avoids that fact: it proves order-independence for arbitrary acyclic federations directly, by bubbling the first registry of one order to the front of the other, every registry it passes being independent of it (`order_independent`, module `FederationOrder.v`). $\square$

5 Compositionality for Free

Because the federated normal forms are a limit and ρ_F is the retraction onto it, a convergent sub-federation behaves as a single effective registry inside a larger one. Split a federation along a topological cut into an upstream block J (closed under sources: every source of a J -registry is itself in J) and the rest K . The *flat* normalizer runs ρ_F over all of $J \cup K$ in one order; the *staged* normalizer runs the sub-federation's normalizer ρ_J to completion, then runs K over the finalized J -block.

Theorem 5.1 (Theorem 2: compositionality). *The flat and staged normalizers agree: normalizing $J \cup K$ in one pass equals normalizing J , then normalizing K over the finalized result. Consequently a verified sub-federation collapses to an effective registry (its valid set L_J), and the combined normalizer factors through it.*

Proof sketch. At the operator level this is the fold-append law: folding J and then continuing with K from the accumulated result equals folding the concatenation, because the accumulator after J is exactly ρ_J 's output and each K -step reads only its own finalized sources. Mechanized as `rhoFold_compositional`, via `rhoF_from_app`, axiom-free. $\square$

This is what lets verification scale: a subsystem is checked once, in isolation, and reused as a black box, with only the seam (the morphisms crossing the cut into K) re-checked. The assume-guarantee refinement makes the black box parametric. A sub-federation may declare *input ports*: shared variables that no internal morphism writes, which an outer morphism may drive once the subsystem is embedded. A seam morphism into a port is verified against $M1/R2$ while the subsystem's interior is trusted from its certificate, and a write to any non-port (sealed) variable is rejected. We treat the port refinement at the paper level here; the operator-level collapse above is the mechanized core.

6 Convergence Certificates Form a Sheaf

The limit of Section 4 is a global object assembled from local ones, the shape of a sheaf. Making that precise both explains why the federation's hypotheses are the right ones and sets up the obstruction of Section 7.

The presheaf. Over a system’s variables, take a region to be a variable subset U , the subsystem on U to be the rules whose footprint lies in U , and the section $F(U)$ to be that subsystem’s normalizer ρ_U together with its convergence verdict. Restriction $F(U) \rightarrow F(U')$ for $U' \subseteq U$ restricts the normalizer. The sheaf question is whether local certificates that agree on overlaps glue to a global one, checkable from the overlap alone.

Gluing is not naive. Agreement on which shared values are *valid* is not enough.

Example 6.1. Let $s \in \{0, 1, 2\}$. Subsystem A has invariant $s \neq 1$ with repair $s := 0$; subsystem B has invariant $s \neq 1$ with repair $s := 2$. Each is confluent, and both have valid set $\{0, 2\}$, so they agree on which values are valid. But in the union, from $s = 1$ the result is 0 (running A first) or 2 (running B first): order-dependent, non-confluent.

The precise gluing condition. The obstruction in Example 6.1 is that A and B disagree as *normalizers* on the shared state ($A : 1 \mapsto 0$, $B : 1 \mapsto 2$). So the correct sense of “agree on the overlap” is that the two certificates restrict to the *same normal-form map* on the shared variables. Then the shared variable has one unambiguous normal form, the union is confluent, and the sections glue: the global normalizer runs each private part plus the agreed map on the overlap. This is checkable on the overlap alone, never on the joint product. Convergence certificates therefore form a sheaf on the site whose overlaps satisfy this agreement.

Two regimes force agreement, and they are the federation’s. The overlap agreement holds in two ways:

- *Single-writer overlap:* only one subsystem writes the shared variable, the other reads it. The writer’s normalizer is the only one on the overlap, so agreement is automatic. This is the authority / $M1$ discipline.
- *Monotone overlap:* both write, but monotonically. The combined repair is a monotone operator with a unique least fixed point (Knaster-Tarski [13]), so the normalizers agree on the glued result. This is the monotone-cycles regime.

Proposition 6.2 ($R1 + R2$ is the gluing axiom). *In a federation, distinct components do not literally share a writable variable; a shared writable variable is a multi-source target whose resolver merges its sources. The resolver hypotheses are exactly the sheaf gluing condition: source-determinacy ($R1$) makes the merged normal form single-valued (confluence on the overlap), and validity-preservation ($R2$) makes it land in the agreed valid set. Hence the federation’s resolver verification is the sheaf gluing check.*

The existing federation design is thus validated by the sheaf structure rather than extended by a new primitive: the machinery that composes overlapping subsystems soundly is already present as $R1/R2$. The gluing counterexample (Example 6.1) is mechanized axiom-free (module `Cohomology.v`); Proposition 6.2 and the sheaf assembly over the full cover are at the paper level.

7 The Cohomological Completion

For two subsystems the gluing condition of Section 6 is a single equality check, so nothing subtle survives. The interesting case is a cycle of three or more subsystems whose pairwise

overlaps each agree yet admit no global section: the classic local-consistency, global-impossibility phenomenon, measured by the first Čech cohomology of the presheaf over the nerve of the overlap cover.

The loop composite. Take a triangle in which A and B share x , B and C share y , C and A share z , the cycle's morphisms fixing each next shared component from the previous normal form. A global section is a shared assignment fixed all the way around, so the shared value must be a fixed point of the loop composite $g = m_{CA} \circ m_{BC} \circ m_{AB}$ on the shared subspace.

Theorem 7.1 (Obstruction, general case). *A global convergent section around a cycle exists iff the cycle's loop composite g has a reachable fixed point, and the witness for its absence is g itself, computable over the shared subspace rather than the product.*

The minimal obstruction has shared values in $\{0, 1\}$, two edges the identity and one the negation, so g is the flip, which has no fixed point (it orbits $0 \mapsto 1 \mapsto 0$): no consistent global assignment exists, and the witness points at the negation edge.

The completion, invertible fragment. Assemble the sheaf on the nerve N of the cover: a vertex per subsystem, an edge per non-empty overlap, a triangle per triple overlap. Write the shared fiber on an overlap as a finite set S ; in the *invertible* fragment (each morphism transports its shared value by a bijection) the identification across edge (i, j) is a bijection g_{ij} in a group $G \leq \text{Sym}(S)$, with the cocycle condition $g_{ik} = g_{jk} \cdot g_{ij}$ on triangles.

H^0 is the convergent states: a global section assigns each subsystem i a shared value s_i with $g_{ij} \cdot s_i = s_j$ on every overlap, which is exactly a consistent global normal form, so $H^0(N, F)$ is the consistent set L_F of Proposition 4.2, and “converges compositionally” is “ H^0 is non-empty.” H^1 is the obstruction: the 0-cochains are gauges ($h_i \in G$ per vertex, a relabeling of subsystem i 's shared normal form), the coboundary is $(dh)_{ij} = h_j \cdot h_i^{-1}$, the 1-cocycles are the transition families satisfying the triangle condition, and H^1 is the 1-cocycles modulo gauge.

Theorem 7.2 (Completion, invertible fragment). *When the shared fiber is a G -torsor, a global section exists iff the transition cocycle $\{g_{ij}\}$ is trivial in H^1 (a coboundary).*

Proof. Fix a base $c \in S$; freeness and transitivity make $g \mapsto g \cdot c$ a bijection $G \rightarrow S$, so each $s_i = h_i \cdot c$ for a unique $h_i \in G$. If $g_{ij} = h_j \cdot h_i^{-1}$, set $s_i = h_i \cdot c$; then $g_{ij} \cdot s_i = h_j \cdot h_i^{-1} \cdot h_i \cdot c = h_j \cdot c = s_j$, a global section. Conversely a section $\{s_i\}$ with $g_{ij} \cdot s_i = s_j$ gives $g_{ij} \cdot h_i = h_j$, so $g_{ij} = h_j \cdot h_i^{-1} = (dh)_{ij}$. $\square$

Mechanized for an arbitrary finite graph and group, with the torsor identified with G by the base point, as `section_iff_coboundary` (module `CohomologyGraph.v`).

H^1 is holonomy, and its generators are a cycle basis. Fix a spanning tree of N and gauge the tree transitions to the identity; the cocycle is then determined by its values on the non-tree edges, one per independent cycle, and each value is the holonomy around that cycle, the loop composite g of Theorem 7.1. So H^1 , as a pointed set, is the holonomies of a cycle basis modulo simultaneous conjugation, its rank is the first Betti number of the nerve, and it is trivial iff every loop composite is the identity. The cycle basis is the minimal obstruction basis: fixing the holonomy on each generator makes H^0 non-empty (Section 9). The generation statement is mechanized: for a spanning tree plus the remaining edges, a global section exists iff every fundamental

cycle has trivial holonomy (`cycle_basis_criterion`, with `sat_iff_trivial_holonomy`), using that sections of a tree are unique up to one constant (`tree_unique`). So is the quotient, on the graph (the nerve’s 1-skeleton, where every labeling is a cocycle): every labeling gauge-fixes to the identity on a spanning tree, its remaining values being the fundamental holonomies (`gauge_fix`, `gauge_fixed_holonomy`); two tree-fixed labelings are cohomologous iff those values are simultaneously conjugate, for any group (`H1_classification`); and the number of generators is $|E| - |V| + 1$ (`betti_number`). The nerve’s triangles add relations that can lower the rank further; those are not modeled.

The two regimes are one picture. The obstruction vanishes in exactly two ways, and they are the two convergence regimes of the base work:

- *Trivial holonomy.* An acyclic nerve has no independent cycles (first Betti number 0, H^1 trivial); more generally trivial loop composites give a coboundary. This is the acyclic / tree route.
- *Monotone fibers.* When S carries a lattice order and the transitions are monotone (not invertible, so outside the torsor picture), Knaster-Tarski gives a least fixed point of the loop composite, so H^0 is non-empty regardless of holonomy. This is the monotone route.

The first kills the cohomological obstruction; the second sidesteps it via order. The dichotomy the base papers state as two separate sufficient conditions is thus one invariant, and its non-vanishing is exactly what the engine rejects.

Scope. The clean cohomological “iff” of Theorem 7.2 is the invertible / torsor fragment. When transitions are non-invertible (a repair that collapses values) the fiber is not a torsor and the transitions are monoid elements, so classical H^1 does not apply; there the obstruction is H^0 directly, via the reachable-fixed-point condition of Theorem 7.1, a dynamical rather than cohomological statement. Mechanized axiom-free: the single-cycle essence of Theorem 7.2, the gluing counterexample, and the negation obstruction (`Cohomology.v`); and Theorem 7.2 itself on an arbitrary graph together with the cycle-basis criterion (`CohomologyGraph.v`), along with H^1 of the graph as a quotient and its rank, for any group. At the paper level: the rank on the nerve as a 2-complex, where the triangle relations can lower it. The operational fixed-point test is implemented (Section 8).

8 The Obstruction Diagnostic

The obstruction of Theorem 7.1 is not only a classification but a computation. The witness for a cycle’s failure to converge is its loop composite g , and g lives on the shared subspace, so it is computed by composing the cycle’s morphisms over that subspace, bounded by it rather than by the product of the components. This is what makes the diagnostic tractable: a cycle through registries with large local state spaces is analyzed on the small shared fiber alone.

Implementation. The reference engine implements this as `Federation.DiagnoseCycle`. It finds a directed cycle in the morphism network, iterates the cycle’s repair from a representative seed on the finite shared space (which must either reach a fixed point or repeat), and reports which, returning the cycle, whether it converged, and, when it did not, the repeating configuration as an orbit witness.

Example 8.1 (Settle versus orbit). Two registries with a shared value in $\{0, 1\}$. If A copies its value to B and B copies back (both edges the identity), the loop composite is the identity: every value is a fixed point, and the diagnostic reports convergence. If instead B copies back the negation, the loop composite is the flip, which orbits $0 \mapsto 1 \mapsto 0$; the diagnostic reports non-convergence and returns that orbit as the witness, pointing at the negation edge.

Reading the verdict. A non-convergent verdict is definitive: an orbit exists, so the cycle cannot converge, and the orbit is a proof. A convergent verdict from a single seed is weaker: it establishes that this seed settles and names the cycle for reference, but does not by itself prove whole-network convergence, since another local state might still oscillate. The diagnostic is therefore a sound refuter and a cycle namer, not a whole-network decision procedure; deciding convergence outright is the exhaustive check the base engine already performs, and the diagnostic explains a rejection rather than replacing that check.

What it delivers. For federating N independently-governed policies, the diagnostic answers two questions the bare convergence verdict does not: can these compose, and, when they cannot, where is the conflict. The answer is legible (it names the offending cycle and exhibits either its orbit or its missing fixed point) and cheap (it composes the cycle’s morphisms and tests for a reachable fixed point over the shared subspace). Combined with Proposition 9.1, it is also constructive: the cycles it flags are exactly the ones any coordinated core must cover.

9 Minimal Coordination

Normalization confluence is a strong-eventual-consistency guarantee: replicas that consume the same events reach the same valid state regardless of order. So the convergence question is a coordination question. A coordination-free, strongly-eventually-consistent implementation exists exactly when a consistent global normal form exists, that is, when H^0 is non-empty, and the obstruction to it is the H^1 class of Section 7. This is the CALM boundary (monotone implies coordination-free) and invariant confluence, now not merely decided but *classified*: the completion says precisely which configurations obstruct, so it turns coordination into a graph problem on the labeled nerve.

Proposition 9.1 (Coordinating a cycle basis suffices). *In the invertible fragment, a federation admits a coordination-free, strongly-eventually-consistent convergent implementation iff $H^1 = 0$. When $H^1 \neq 0$, imposing a coordination boundary (a total order, or a single writer) on the shared variable along a cycle basis trivializes the obstruction, yielding a convergent implementation that is strongly consistent only on those loops and coordination-free everywhere else.*

Proof sketch. Coordinating a loop forces its shared variable to a single agreed value, the single-writer overlap of Section 6, so that loop’s transition becomes a coboundary and its holonomy generator dies. Coordinating a cycle basis kills every generator, leaving the trivial class, so H^0 becomes non-empty by Theorem 7.2; off the coordinated loops the transitions are unchanged and stay coordination-free. $\square$

Mechanized: a spanning tree always carries a section, so coordinating every non-tree edge suffices (`tree_has_section`); and relative to a fixed spanning tree, the edges that must be coordinated are exactly the unbalanced fundamental edges (`keep_balanced_suffices`, `unbalanced_blocks`), with a $\mathbb{Z}/2$ triangle discharging every hypothesis.

So a federation decomposes into a coordination-free majority (eventual consistency) and a coordinated core (strong consistency). For a distributed deployment this reads: put consensus only on the obstructing cycles, run everything else coordination-free. It refines the global coordination verdict of CALM [5, 6] and invariant confluence [7] from a single yes-or-no into a localized, mixed-consistency partition.

The minimum coordination, and its complexity. A cycle basis suffices but is not minimal, since one coordinated edge can break several obstructing cycles at once. The exact minimum is the *group feedback edge set* number of the labeled nerve: delete the fewest shared variables so every residual cycle is balanced. This resolves its complexity. It is NP-hard even in the abelian case, since for $G = \mathbb{Z}/2$ it is edge bipartization, the complement of Max Cut. Under the Unique Games Conjecture it has no constant-factor approximation ($O(\sqrt{\log n})$, the Min-Uncut bound, is the best known). But it is fixed-parameter tractable in the size of the coordinated core, which is the operative regime since one wants that core small, and polynomial on planar nerves (planar Max Cut) and on edge-disjoint obstructions (one hit per cycle, any G). A correct, polynomial, size- $\leq b$ coordination is always available: coordinate the orbiting fundamental cycles of any spanning tree.

Non-commutativity raises the floor; it never lowers it. The minimum is never below what the abelianized holonomy suggests, and can be strictly above. A section after coordinating F needs $\pi_1(N \setminus F) \subseteq \ker \rho \subseteq \ker(\text{ab} \circ \rho)$, so every feasible F is feasible for the abelianized problem and $\min_G \geq \min_{G^{\text{ab}}}$. The inequality is strict: on the theta graph with holonomies (0 1 2) and (0 1) in S_3 , the true minimum is 2, while the sign-abelianized count is 1 because the 3-cycle is even and invisible to the sign. The consequence is a soundness warning: sizing coordination by any abelianized invariant under-provisions, wrongly declaring coordination-free some cyclic federations that in fact need to coordinate an even-permutation relabeling loop. This separation is machine-checked end to end: the crux permutation facts (`Cohomology.v`) and, stated over global sections of the S_3 torsor, the counts $\min_G = 2$ and $\min_{G^{\text{ab}}} = 1$, the fact that sign is the abelianization homomorphism, and the functoriality that every S_3 section induces an abelian one (`CohomologyMin.v`, `theta_separation`).

Which federations inhabit the cohomological fragment. A morphism is invertible on the shared fiber exactly when it transports the shared value by a bijection (a lossless relabeling: a copy, a permutation, a bijective unit or schema map), and non-invertible when it collapses information (a clamp, a reset, a most-restrictive-wins merge, which includes every multi-source resolver). The clean cohomological classification is the invertible case; acyclic federations have nothing to coordinate, monotone cycles converge by the least fixed point, and only invertible cycles with non-trivial holonomy require the coordinated core. In the general non-invertible case the decomposition still holds (a cycle basis suffices, by the fixed-point form of Theorem 7.1), with the same group-feedback-set minimum.

From rejection to repair. The base engine rejects a non-monotone cyclic federation: it cannot converge coordination-free, and the build fails. Proposition 9.1 upgrades that blunt verdict into an actionable one: rather than reject, accept the federation with a coordination boundary on a cycle basis, which the diagnostic of Section 8 identifies. This turns “these policies do not compose” into “these policies compose if you coordinate exactly these loops.”

Boundary. This locates where coordination is needed for convergence, assuming the delivery layer gives each replica the same event set (normalization confluence is the convergence layer, not the transport). It is clean in the invertible fragment; the general non-invertible case reduces to the loop-composite fixed-point condition, where coordinating a loop still means forcing its shared variable to an agreed value but the minimal-basis count is not a cohomological rank.

10 Conclusion

Federated convergence has a structure, and naming it answers both a theoretical and a practical question. Theoretically, a federation’s convergent states are a limit and its normalizer is the retraction onto that limit, so compositionality is a corollary rather than a separate theorem; the convergence certificates form a sheaf whose gluing axiom is exactly the federation’s already-certified $R1/R2$; and the obstruction to a global convergent state is cohomological, with H^0 the convergent states and H^1 the holonomy of the morphism cycles. The two convergence regimes of the base work, acyclicity and monotone repair, are then one invariant: the first makes H^1 vanish, the second sidesteps it by order.

Practically, the same invariant is computable and actionable. Its witness is a cycle’s loop composite over the shared subspace, implemented as a diagnostic that names an obstructing cycle and exhibits its orbit; and when the obstruction is non-zero, coordinating a cycle basis suffices to converge. This localizes the global coordination verdict of CALM and invariant confluence into a mixed-consistency partition, and it turns a rejection into a constructive “coordinate exactly these loops.” The exact minimum coordination is the group feedback edge set number, NP-hard even in the abelian case but fixed-parameter tractable in the core size and polynomial on planar or edge-disjoint nerves, and it is never below the abelian count.

The structural core, the limit, the retraction for arbitrary acyclic federations, compositionality, and full order-independence, is mechanized axiom-free, and so is the cohomological layer: the completion on an arbitrary graph, the cycle-basis criterion, the sufficiency of coordinating a spanning tree’s complement, H^1 of the graph as tuples of fundamental holonomies modulo simultaneous conjugation with rank $|E| - |V| + 1$, and the S_3 separation. What is deliberately left at the paper level is scoped plainly: the clean cohomological classification is the invertible (lossless-relabeling) fragment, the general case reducing to the loop-composite fixed-point condition that the diagnostic already computes; the rank on the nerve as a 2-complex is not mechanized; and the complexity of the minimum is cited from the group feedback set literature. The open problems are a cohomological account of the non-invertible case and the classification on the full 2-dimensional nerve.

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